

Business Strategy, Executive Compensation, and Firm Performance

Abstract

This study investigates the influence of business strategy on the relationship between executive compensation and firm performance. Using cluster analyzes to classify a firm's business strategy, we predict and find that performance-linked compensation and managerial share ownership are more effective for product differentiators than for cost leaders. The results are consistent with the view that managers of product differentiation firms are more willing to take risks and to make a trade-off between pay security and share ownership to benefit from anticipated future performance. We also find that, contrary to our prediction, current payment of a long-term incentive plan is less relevant for product differentiators than for cost leaders. One plausible explanation for this finding is that long-term incentive payouts reflect past performance that is less persistent for firms seeking innovation and differentiation. This study contributes to the existing literature on executive compensation by recognizing that the effects of executive compensation on performance vary systematically across business strategies.

Key words: Business strategy, executive compensation, cluster analysis, performance

Business Strategy, Executive Compensation, and Firm Performance

1. Introduction

The relationship between executive compensation and firm performance has attracted considerable attention both from researchers and practitioners. From an agency theory perspective, performance-linked compensation (PLC) provides incentives for executives to take actions for the best interests of shareholders (Jensen and Meckling, 1976; Fama and Jensen, 1983). Agency theory proposes that PLC motivates executives to make a trade off between the risks imposed on them and incentives associated with certain performance measures. Linking executive compensation with certain performance measures enables firms to observe which actions and decisions were taken by the executives (Indjejikian, 1999). Therefore, it is not surprising that many firms link their compensation packages with firm objectives. Lublin (2006), for example, reports that about 30 per cent of major U.S. companies link their executive compensation packages to firm performance.

Despite the widely held premise that executive compensation is an important mechanism to align the interests of executives with those of the shareholders, previous empirical studies have reported a weak and even insignificant relationship between executive compensation and firm performance (e.g., Jensen and Murphy, 1990; Kerr and Bettis, 1987). Barkema and Gomez-Mejia (1998) suggest that researchers should exert greater efforts to examine the influence of a competitive strategy on the design of executive compensation, and how the fit between a firm's chosen strategy and the structure of executive compensation affects firm performance; firms tend to use different criteria consistent with their chosen strategy to compensate their executives. Rajagopalan and Finkelstein (1992) argue that executive compensation will affect performance positively only if it is designed to be congruent with a firm's chosen strategy.

The objective of this study is to investigate the influence of business strategy on the relationship between executive compensation and performance. We argue that PLC, managerial share ownership, and long-term compensation plans are more effective for product differentiation firms than for cost leadership firms. Using data from Compustat S&P 500 and ExecuComp databases, we found that both PLC and managerial share ownership are more effective for product differentiation firms than for cost leadership firms. However, contradictory to our expectation, we found that a long-term incentive plan payout is negatively related to current-year performance and that the relationship is more negative for product differentiation firms than for cost leadership firms. Overall, the results indicate that the impact of executive compensation on performance varies depending upon a firm's business strategy.

The remainder of this paper proceeds as follows. The next section reviews the related literature and presents the study's hypotheses. Section three describes the research method, and section four details the data analyses and the results of statistical tests. The final section discusses the study's major findings and limitations, as well as its implications for future research in this area.

Related literature and Hypotheses

Agency theory is one of the dominant theoretical perspectives used in the accounting literature to analyze the relationship between executive compensation and performance. According to agency theory, the separation of ownership and management creates an agency problem (Jensen and Meckling 1976).

An agency problem is characterized by the presence of diverging interests, differences in risk aversion, different decision horizons, and an information asymmetry between the principal

and the agent. Given the diverging interests of the agent and principal and the assumption of a “selfish man,” agency theory predicts that managers who have an information advantage will maximize their own interest instead of that of the principal. This phenomenon is defined as “moral hazard” in the the economics literature. If shareholders have complete information regarding the managers’ activities, they could design a contract specifying and enforcing managerial actions (Jensen and Meckling, 1976).

However, managerial actions are not perfectly observable by shareholders. In such a case, agency theory predicts that firms will design a compensation policy to give managers an incentive to select and implement actions that maximize shareholders’ wealth. One of the solutions for mitigating the moral hazard is to introduce an incentive plan, which explicitly links the agent’s compensation to the organization’s performance. Firms often specify the performance targets, and measure the extent to which these targets have been met through the agent’s efforts, and reward the agents accordingly. Schwab (1973) suggests that the type of system used by a firm to compensate its employees is an important variable that will influence managers’ motivation to perform. Indeed, in their comprehensive study of executive remunerations, Jensen and Murphy (2004) recommend, among other things, that firms should “design bonus plans with ‘linear’ pay-performance relationship” (p. 12)¹. Ironically, in their previous study, the same authors expressed disappointment when they found a low pay-for-performance sensitivity. They concluded that “our results are inconsistent with the implications of formal agency models of optimal contracting” (Jensen and Murphy, 1990, p. 227). In a follow-up study, Garen (1994) also found that the pay-for-performance sensitivity is quite low.

¹ This recommendation is consistent with the prescription of expectancy theory. When managers perceive that there is a link between their efforts and their compensations, they will be motivated to exert higher efforts for superior performance (Lawler, 1983). In a similar vein, Lawler (1971) suggests that executive compensations will have higher valence when they are attached to performance that requires effort.

He concluded that although the principal-agent perspectives have an important effect on executive compensation, other factors must be considered when investigating the relationship between incentive-based compensation and performance. Barkema and Gomez-Mejia (1998) assert that the “failure to identify a robust relationship between top management compensation and firm performance has led scholars into a blind alley” (p. 135).

The weak results reported by prior studies on the relationship between executive compensation and performance may be due, in part, to the researchers’ approach. Most of the researchers who conducted these studies used the universal approach, which examines the direct or main effects of executive compensation on performance. Balkin and Gomez-Mejia (1999) note that these prior studies overlooked the effects of a firm’s business strategy and contend that this may account for their weak and often insignificant results. Balkin and Gomez-Mejia (1999) argue that studies examining the relationship between executive compensation and performance should consider the firms’ business strategy for three reasons: 1) different strategies require different pay policies, (2) the relative effectiveness of different pay policies varies across strategies, and (3) the misfit between business strategies and compensation policies will negatively affect performance. This is the challenge that is explored by the study presented in this paper.

Business strategy, performance-linked compensation and performance

In general, there are two broad categories of business strategy. Porter (1985) develops a framework that outlines how firms might choose a business strategy in order to compete effectively. He argues that a firm must choose between competing as the lowest-cost producer in its industry (i.e., a cost leadership strategy) or competing by providing unique products in terms of quality, physical characteristics, or product-related services (i.e., a product differentiation

strategy)². In addition, he emphasizes that the essence of a firm's business strategy is its ability to deliberately choose a set of activities which will deliver a unique mix of values to its customers (Porter, 1996).

The existing literature contains some discussion of why the relationship between executive compensation and performance depends on a firm's choice of strategy. There are trade-offs in using performance-related compensation to alleviate the agency costs. Performance-related compensation is based on observable outcomes of the firm, not on the direct observation of managerial efforts. Such a plan provides an incentive for the agent to work harder but at the same time transfers risk onto a risk-averse agent. Therefore, we argue that performance-related compensation is a good fit for product differentiators, where managers' efforts in creating unique products are less observable, and risk-taking behaviors are more likely to get rewarded.

First, product differentiation firms emphasize customer satisfaction by offering high-quality products and services, specialized design features, fast and reliable delivery, and effective post-sales support (Lynn, 1994). With their primary focus on uniqueness and exclusivity, product-differentiation firms tend to engage more in product innovation (Porter, 1980) and therefore need to invest heavily in research and development activities (Miller, 1987) than low-cost firms do. Miller (1987) argues that investment in research and development activities increases a firm's ability to produce new and better products and helps it to keep up with innovations of competitors and the changing environment. Biggadike (1979) argues that firms with a strong emphasis on new products will face high uncertainty, since they are involved in riskier activities and are betting on products that have not yet crystallized. These firms require

² Porter (1985) asserts that firms pursuing a cost-leadership strategy do not imply that they can ignore quality, service, features, or other bases for differentiation. Similarly, firms pursuing a product differentiation strategy cannot ignore costs. However, the primary strategic target for cost leaders is efficiency, while the primary strategic target for product differentiators is innovativeness.

managers to take risks and to have greater tolerance for ambiguity (Gupta and Govindarajan, 1984). These firms also tend to attract more risk-taking managers who are willing to spend extra efforts to earn higher rewards in favor of pay security (fixed salary) and immediate rewards (Balkin and Gomez-Mejia, 1987). Managers under a PLC policy are invited to be risk-takers and share in the anticipated profits with the owners (Balkin and Gomez-Mejia, 1987). The advantage of a PLC policy is that employees allow firms some leeway; the employers pay the employees when they are in the best financial position to do so, or when the financial or other strategic objectives have been reached. As a result, product differentiation firms can secure greater flexibility to invest heavily in research and development activities and marketing expenses (instead of paying a fixed amount of compensation). Furthermore, Welbourne et al., (1995) argue that it is difficult to determine the appropriateness of managers' decisions in riskier firms, and that performance-linked compensation may enhance managers' self-monitoring.

Second, product differentiation firms are expected to find PLC to be more effective than cost-leadership firms. Burton et al. (2002) argue that cost-leadership firms will gain a competitive advantage by planning their activities reasonably well to realize efficiency. Porter (1985) suggests that to increase efficiency cost-leadership firms should produce standard products and employ standardized operating procedures. Although these practices reduce flexibility in responding to changes in technology, customer demand, and competition, they increase efficiency by narrowing the range of tasks and by making activities more routine (Miller, 1987). Cost-leadership firms tend to have a well-defined mean-end relationship, and their evaluation process tends to be routine and mechanistic. In such an environment, firms should emphasize fixed pay established through a formal job evaluation process (Balkin and Gomez-Mejia, 1987). By contrast, product differentiation firms tend to invest heavily in research

and development activities to generate new products or processes. Bublitz and Ettredge (1989) contend that without proper monitoring and incentives, managers tend to take advantage of the highly discretionary nature of R&D expenditures. Hence, linking managers' incentives with firm performance will discourage managers to engage in opportunistic behaviors, resulting in better outcomes of the R&D activities.

The preceding discussions indicate that PLC should be more effective for product-differentiation firms than for cost-leadership firms since firms' emphasis on risk-taking and high rewards fits in with the product-differentiation strategy that involves high-risk-and-high-return activities. Therefore, the following hypotheses will be tested:

Hypothesis 1: Performance-linked compensation will be more effective for product differentiation firms than for cost leadership firms.

Business strategy, managerial share ownership and performance

Managerial share ownership can have a significant impact on the governance of a firm because it gives managers the right to vote on important decisions and provides an incentive for them to act in the best interest of shareholders. Jensen (1993) argues that managerial-share ownership reduces the likelihood that managers will behave opportunistically since it aligns their interests with shareholders' interests. When managers hold more equity in a firm, their incentive to work hard increases since they will share a bigger portion of the proceeds from the firm's success (Warfield et al., 1995). Managerial share ownership will also discourage managers from engaging in shirking and perquisite taking (Cheng et al., 2003; Chow et al., 1997). Indeed, Jensen & Meckling (1976) suggest that as managers' fraction of the equity increases, their fractional claim on the outcomes rises, which will tend to encourage them to

appropriate smaller amounts of the firms' resources in the form of perquisites. Hence, the relationship between managerial share ownership and performance will be positive.

Previous studies (e.g., Warfield et al., 1995; Hutchinson and Gul, 2004; Gul et al., 2002) have shown that share ownership motivates managers to use discretion when they invest in value-maximizing activities. For example, Warfield et al. (1995) used data from Standard and Poor's Compustat database in finding that the correlation between stock returns and accounting earnings is positively affected by managerial share ownership. In a similar vein, Hutchinson and Gul (2004) used data from 500 firms listed on the Australian Stock Exchange to show that managerial share ownership mitigates the negative association between a firm's investment opportunities and performance.

In keeping with Jensen (1993), we also argue that managerial share ownership aligns the interests of managers and shareholders. However, this study goes one step further and argues that business strategy will influence the relationship between share ownership and performance. As discussed in the previous section, product differentiation firms are riskier than cost leadership firms. Demsetz and Lehn (1985) contend that the scope for moral hazard is greater for managers of riskier firms who tend to consume more perquisites to compensate for the higher risks they are assuming (Balkin and Gomez-Mejia, 1987). Therefore, it is reasonable to assume that owners will provide managers with greater ownership stakes to align incentives (Himmelberg et al., 1999).

Hence, we argue that managerial share ownership will be more effective for product differentiation firms than for cost leadership firms. First, managers' actions are less observable for product differentiators than for cost leaders (Govindarajan and Fisher, 1990). With standardized products and production processes, the means-and-ends relationship for

cost leaders is more observable. The owners of cost leadership firms can monitor their managers effectively through a formal job-evaluation process. Under the optimal contracting regime, the owners of cost leadership firms will be able to monitor their managers' actions reasonably well resulting in higher performance as managers will not be able to divert resources for perquisites (Himmelberg et al., 1999). For product differentiators, the means-and-ends relationship will be unclear, and monitoring will be more difficult. Therefore, the owners of product differentiation firms need to offer greater ownership stakes for their managers to align incentives.

Second, R&D firms tend to have a higher fraction of their assets in the form of intangibles such as R&D projects and patents. The owners of product differentiation firms will require managerial ownership to align incentives because intangible assets are harder to monitor and are therefore subject to managerial discretion (Himmelberg et al., 1999).

Third, product differentiation firms tend to enjoy quasi-monopoly from their innovative products. Williamson (1963) argues that managerial behavior toward discretionary spending is contingent upon the conditions of competition. He contends that vigorous competition will prevent managers from manipulating the activities of the firm in order to pursue their own goals while an absence of this condition will permit managers to pursue their own goals. On the other hand, since cost leaders tend to produce commodity products that are subject to severe competition, this market condition will act as a monitoring mechanism to discipline managers.

Previous discussions suggest that the effectiveness of managerial share ownership to align incentives will be different across different strategies. Specifically, the following hypothesis will be tested:

Hypothesis 2: Managerial share ownership will be more effective for product differentiation firms than for cost leadership firms.

Business strategy, long-term incentives and performance

Product differentiation firms tend to invest heavily in R&D activities in order to increase their innovative capability and enhance their ability to keep up with their competitors' innovations (Miller, 1987). The outcomes of these activities are usually long term in nature (Govindarajan and Fisher, 1990). The key success factors for product differentiators include creative skill, strong basic research, product engineering (Porter, 1980), and sufficient time to succeed in accomplishing these factors (Ouchi, 1979). Therefore, product differentiation firms need to provide incentives for their managers to focus on the long-term success of the firm. Furthermore, product differentiators tend to attract younger, more risk-taking managers (Ettlie, 1983) who are usually more willing to make a trade-off between job security and immediate rewards for future success (Balkin & Gomez-Mejia, 1985). In summary, a long-term incentive plan is a good fit with a product differentiation strategy. Specifically, the following hypothesis will be tested:

Hypothesis 3: Long-term incentives will be more effective for product differentiation firms than for cost leadership firms.

3. Research Methodology

3.1 Sample selection

The initial sample includes all firms listed in the Compustat S&P 500 database for the period 2000-2005. This period was chosen in order to reduce the incomparability of executive compensation data before and after the Enron scandal in 1999, which has caused increased

scrutiny of executive compensation. Information about executive compensation is collected from the S&P ExecuComp Database. To be included in the sample, a firm has to report research and development expenditures, net income, cost of goods sold, total sales revenues, market return, and total assets in the Compustat S&P 500 database. In addition, the following variables must be available in the ExecuComp Database: total compensation (TDC1), total salary (SALARY), percentage of the company's shares owned by the named executive officer (SHROWNPC), long-term incentive plan paid (LTIP), number of director meetings (NUMMTGS), total meeting payout (NUMMTGS*DIRMTGFEE), and the number of years credited service the named executive has under the firm's pension plan (RETYRS).

[Insert Table 1 here]

Table 1 summarizes the sample selection. The initial sample consists of 515 firms. Two hundred and fifty seven firms were eliminated because we lacked the data needed to perform the cluster analyses. This elimination reduces the total number of firms to 234 or 1,404 firm-year observations. In addition, 655 firm-year observations were eliminated for a lack of the data needed for the statistical analyses. The final sample consists of 749 firm-year observations (258 and 491 firm-year observations for product differentiators and cost leaders, respectively).

3.2 Research Design

To investigate the effects of business strategy on the relationship between executive compensation and performance, we use the following moderated regression model:

$$\begin{aligned} PERFORM_{it} = & \gamma_0 + \gamma_1 PLC_{it} + \gamma_2 MSOWN_{it} + \gamma_3 PLTIP_{it} + \gamma_4 STRA_{it} + \gamma_5 STRA_{it} * PLC_{it} + \gamma_6 \\ & STRA_{it} * MSOWN_{it} + \gamma_7 STRA_{it} * SALARY_{it} + \gamma_8 STRA_{it} * PLTIP_{it} + \gamma_9 MTGS_{it} + \\ & \gamma_{10} MTGSFEE_{it} + \gamma_{11} RETYRS_{it} + \gamma_{12} SIZE_{it} + \varepsilon \end{aligned}$$

Where:

$PERFORM_{it}$	= Performance of firm i in year t determined by the one-year total return to shareholders, including the monthly reinvestment of dividends (ExecuComp data item TRS1YR).
PLC_{it}	= Performance-linked compensation of firm i in year t determined by the percentage of incentive-based compensation to total compensation.
$MSOWN_{it}$	= Percentage of the firm's shares owned by the named executive officers.
$SALARY_{it}$	= Total base salary earned by the named executive officers as a percentage of their total compensation.
$PLTIP_{it}$	= The amount paid out to the executives under the firm's long-term incentive plan as a percentage of total compensation.
$STRA_{it}$	= An indicator equal to 1 for cost leadership firms and 0 for product differentiation firms.
$MTGS_{it}$	= Number of board meetings held during year t for company i .
$MTGSFEE_{it}$	= Amount paid to directors for attending the board meetings.
$RETYRS_{it}$	= Total number of years of credited services the named executive officers have under the firm's pension plan
$SIZE_{it}$	= Firm size as measured by the total assets of firm i in year t .

Note that we include number of board meetings (MTGS), amount of fees paid to directors for attending the board meetings (MTGSFEE), total number of years of the officers' tenure (RETYRS), and firm size (SIZE) as control variables. More frequent board meetings mean boards have more opportunity to scrutinize management decisions. Core et al., (1999) found that more effective monitoring activities by the boards increase firm performance. In a similar vein, the amount of fees paid to the directors to attend board meetings will provide incentives to meet

more frequently and to provide more regular reviews of management's activities resulting in less opportunity for managers to pursue their own interests (Shleifer and Vishny, 1997). Hence, we expect that both MTGS and MTGSFEE will have a positive relationship with performance. Directors' tenure represents the stability of the board. Crutchley et al. (2002) found that greater board stability is associated with improved performance. Therefore, we predict that RETYRS will have a positive relationship with performance. Previous studies have also shown that firm size may affect performance (e.g., Frank and Goyal, 2003; Ramaswamy, 2001). Therefore, this study includes SIZE to ensure that control for the potential effect of firm size on the results of this study. We expect that firm size will have a positive effect on performance.

4. Data Analyses and Results

This study begins the analyses by conducting a hierarchical cluster analysis to classify the sample firms into strategic archetypes using Porter's (1985) framework. Following Gani and Jermias (2006), this study uses three classificatory variables: R&D intensity (as measured by the ratio of Research and Development Expenses to Total Sales), Asset Utilization Efficiency (as measured by the ratio of Total Sales to Total Assets), and Premium Price Capability (as measured by the ratio of Gross Margin to Total Sales). Gani and Jermias (2006) argue that these variables reflect two critical dimensions of business strategy in Porter's (1985) model, "the search for new or unique products enabling firms to charge premium price" and "asset utilization efficiency." Product differentiation firms are expected to invest heavily in R&D activities enabling them to produce innovative products and services. Successful innovation will give the firms a quasi-monopoly allowing them to charge premium prices. Therefore, product differentiation firms are expected to have higher ratios of R&D intensity and premium price capability than cost leadership firms. By contrast, cost leadership firms focus on achieving

efficient operations through economies of scale, tight cost and overhead control, and cost minimization in areas like R&D, sales force, and advertising (Porter, 1980). Consequently, cost leadership firms are expected to have higher ratios of asset utilization efficiency than product differentiation firms.

The icicle plots, agglomeration coefficients, and the dendrogram produced by the hierarchical cluster analysis were analyzed. Table 2 reports the results of these analyses and indicate that there are two distinct clusters that show expected characteristics with respect to the classificatory variables. A t-test indicates that cluster one has significantly higher ratios of R&D intensity ($t = 8.159$, $p < 0.001$) and premium price capability ($t = 5.870$, $p < 0.001$), but has significantly lower ratios of asset utilization efficiency ($t = -35.353$, $p < 0.001$) as compared to those of cluster two. Therefore, cluster one was defined as a product differentiation group and cluster two as a cost leadership group.

[INSERT TABLE 2 ABOUT HERE]

4.1. Descriptive Statistics and Correlations

Table 3 reports the descriptive statistics for the dependent, the independent, and the control variables. The average shareholder return of the sampled firms is 40.44 % with a minimum of -25.00 % and a maximum of 6,150.79 %. The average performance-linked compensation as a percentage of total compensation is 75.68 % with a minimum of 29.16% and a maximum of 97.86%. The average number of shares owned by the named executive officers is 0.82% with a minimum of 0.00% and a maximum of 38.62%. The firms pay their executives, on average, an 8.88 % long-term incentive plan, with a minimum of 0.00 and a maximum of 72.42%. Sixty-four percent of the sampled firms are categorized as cost leaders, and thirty-six percent are categorized as product differentiators. On average, the boards meet 7.44 time with a

minimum of 2 and a maximum of 30 meetings. The average meeting fee of the directors is \$7,700 with a minimum of \$0.00 and a maximum of \$60,000. The average total number of years that the named executive officers have under the firms' pension plan is 55.62 with a minimum of zero and a maximum of 208. The firms' average size as measured by the firms' total assets is US \$ 18,690.27 million with a minimum of US \$ 114.98 million and a maximum of US \$ 750.51 billion.

[INSERT TABLE 3 ABOUT HERE]

Table 4 reveals the Pearson (bottom part) and Spearman (top part) correlations among variables used in this study. The results indicate that both the Pearson and Spearman correlations are qualitatively similar, and therefore we will discuss only the Pearson correlations with respect to the test variables. As expected, PLC and MSOWN are significantly and positively related to PERFORM. PLC is significantly and negatively related to MSOWN indicating that the ratio of performance-linked compensation to total compensation decreases as managers hold more of the firms' shares. The positive and significant correlation between PLC and PLTIP suggests that firms raise their long-term incentive plan as the performance-linked compensation increases. The negative and significant correlation between PLC and STRA indicates that cost leadership firms use less performance-linked compensation than product differentiation firms.

[INSERT TABLE 4 ABOUT HERE]

4.2. Hypothesis Testing

Table 5 presents the results of the panel-data regression analyses³. Regression 1 (with the interaction terms) shows the interactive effects of business strategy on the relationship

³ The analyses were performed using the xtreg command in STATA.

between executive compensation (PLC, MSOWN, PLTIP) and performance. Regression 2 (without the interaction terms) reports the main effects of the independent variables on performance. Given that the interaction terms are all statistically significant, regression 1 provides a better picture of the relationship between executive compensation and performance (Cohen et al., 2003).

Collinearity diagnostic tests were also performed on the independent variables used in this study. The variance inflation factors (VIF) are reported for each variable to demonstrate the stability of the regression model. The VIF are negatively related to the stability of the regression model. Column (5) in Table 5 indicates that none of the independent variables have VIF values larger than 10 suggesting that multicollinearity should not be a problem in interpreting the regression results⁴.

Column (3) in Table 5 reports the results of the panel-data regression model. Hypothesis H1 predicts that performance-linked compensation will be more effective for product differentiation firms than for cost leadership firms. The hypothesis concerns the incremental effects of STRA on the relationship between PLC and PERFORM as indicated by the interaction of STRA and PLC. The results indicate that the coefficient on STRA*PLC is negative and statistically significant ($b = -1.03$, $p < 0.05$). Conditioned on the STRA being equal to 1 for cost leaders and 0 for product differentiators, the relationship between PLC and PERFORM for product differentiators is represented by the coefficient on PLC. For cost leaders, the relationship is represented by the sum of the coefficients on PLC and STRA*PLC. The results therefore indicate that performance-linked compensation is more effective for product differentiation firms

⁴ Dielman (2001) proposes that any individual VIF larger than 10 indicates that multicollinearity may be influencing the least-squares estimates of the regression coefficients.

(PLC = 1.71) than for cost leaders ($1.71 \text{ PLC} - 1.03 \text{ STRA} * \text{PLC} = 0.68$). These results are consistent with hypothesis H1.

Hypothesis H2 expects that managerial share ownership will be more effective for product differentiation firms than for cost leadership firms. This hypothesis deals with the incremental effects of STRA on the relationship between MSOWN and PERFORM as indicated by the interaction between this variable with MSOWN. Consistent with the expectation, the coefficient on STRA*MSOWN is significantly negative ($b = -3.28, p < 0.05$). Conditioned on the STRA being equal to 1 for cost leaders and 0 for product differentiators, the relationship between MSOWN and PERFORM for product differentiators is represented by the coefficient on MSOWN. For cost leaders, the relationship is represented by the sum of the coefficients on MSOWN and STRA*MSOWN. These results, therefore, suggest that managerial share ownership is more effective for product differentiation firms ($\text{MSOWN} = 3.04$) than for cost leaders ($3.04 \text{ MSOWN} - 3.28 \text{ STRA} * \text{MSOWN} = -0.24$). These results confirm hypothesis H2.

Hypothesis H3 predicts that a long-term incentive plan will be more effective for product differentiation firms than for cost leadership firms. This hypothesis is concerned with the incremental effects of STRA on the relationship between PLTIP and PERFORM as indicated by the interaction between this variable and PLTIP. Contrary to our prediction, the coefficient on STRA*PLTIP is significantly positive ($b = -1.49, p < 0.001$). Conditioned on the STRA being equal to 1 for cost leaders and 0 for product differentiators, the relationship between PLTIP and PERFORM for product differentiators is represented by the coefficient on PLTIP. For cost leaders, the

relationship is represented by the sum of the coefficients on PLTIP and STRA*PLTIP. These results, therefore, suggest that a long-term incentive plan is less effective for product differentiation firms ($PLTIP = -0.88$) than for cost leaders ($-0.88 PLTIP + 1.49 STRA*PLTIP = 0.61$). These results contradict hypothesis H3.

With respect to the control variable, the results indicate that firm performance is negatively related to the number of board meetings ($b = -1.83, p < 0.10$) and also firm size ($b = -8.10, p < 0.001$).

[INSERT TABLE 5 ABOUT HERE]

The moderating effects of business strategy on the relationship between executive compensation and performance can be better understood by using graphical terms. Figure 1 depicts the predicted values of performance for product differentiation firms and cost leadership firms as a function of performance-linked compensation; these values are based on the regression estimates reported in Table 5. The pattern of relationship between PLC and PERFORM differs depending on a firm's strategy. Although the impact of PLC on PERFORM is positive for both product differentiation firms and cost leadership firms, the relationship is stronger for product differentiators than for cost leaders.

[Insert Figure 1 here]

Figure 2 shows the predicted values of performance for product differentiation firms and cost leadership firms as a function of managerial share ownership; these values are based on the regression estimates reported in Table 5. The pattern of relationship between MSOWN and PERFORM differs depending on a firm's strategy. The impact of MSOWN on PERFORM is positive for product differentiation firms but negative for cost leadership firms.

[Insert Figure 2 here]

Figure 3 illustrates predicted values of performance for product differentiation firms and cost leadership firms as a function of a long-term incentive plan; these values are based on the regression estimates reported in Table 5. The pattern of relationship between PLTIP and PERFORM differs depending on a firm's strategy. The impact of PLTIP on PERFORM is negative for product differentiation firms but positive for cost leadership firms.

[Insert Figure 3 here]

Note that the dependent variable in our model is the firms' current performance (year t). To investigate whether the long-term incentive plan (PLTIP) affects firms' future performance, we regress this variable on firms' three-year shareholder return (from year $t-3$ to year t) and five-year shareholder returns (from year $t-5$ to year t)⁵. The results are presented in Table 6 Panels A and B.

[Insert Table 6 here]

Table 6, Panels A and B reveal that the relationships between long-term incentive plans and long-term performance are positive and statistically significant for both three-year ($b = 0.45$, $p < 0.01$) and five-year performance ($b = 0.22$, $p < 0.01$). These results support the view that a long-term incentive plan provides incentives for managers to focus on activities that drive long-term performance. The interactions between STRA and PLTIP, however, are not statistically significant for both specifications.

5. Discussions, Limitations, and Directions for Future Research

In this paper, we investigate the effects of business strategy on the relationship between executive compensation and performance. As predicted, we found that performance-linked compensation is more effective for product differentiation firms than for cost leadership firms.

⁵ We use ExecuComp items TRS3YR and TRS5YR as measures of three- and five-year performance respectively. These two data items measure the total return to shareholders by the sum of capital appreciation and the dividend yields, based on the assumption that dividends distributed will be reinvested.

These results are consistent with the view that riskier firms benefit more from performance-linked compensation because these firms will pay their employees only when they are in the best financial position to do so (i.e., they have reached their strategic objectives) (Balkin and Gomex-Mejia, 1987). Furthermore, the results support the view that performance-linked compensation is more effective for firms characterized with a high degree of managerial discretion such as those with heavy investments in R&D activities since this compensation system imposes managers' self-monitoring (Welbourne et al., 1995). We also found that managerial share ownership is more effective for product differentiation firms than for cost leadership firms. The results even show that for cost leadership firms, managerial share ownership is negatively related to performance. These results imply that when managers' actions can be easily observed (i.e., due to standardized products and production processes), cost leadership firms do not need to provide share ownership to their managers to align incentives (Himmelberg et al., 1999). Contrary to our expectation, the result indicates that a long-term incentive plan is less effective for product differentiation firms than for cost leadership firms. There are two plausible explanations for this contradictory result. First, the market may penalize companies that make cash payments of long-term incentive plans because the payment will reduce the firms' resources to invest in future R&D projects. Second, the long-term incentive payment might indicate good performance in the past, but not necessarily imply good performance in the current or future periods. This is particularly true for product differentiators in which managers' efforts that have created successful products in the past might not be applied to the current and future periods. By contrast, cost leaders have limited lines of products with standardized production processes. As such, cost-cutting techniques that have lead to better performance in the past can also be used to reduce

costs in the subsequent periods. Therefore, the relationship between a long-term incentive plan payout and current-period performance might be weaker for differentiators than for cost leaders.

The results of this study, however, should be interpreted in light of two limitations. First, this study uses data from US firms. Further research is required in order to determine whether the results reported here can be extended to other countries. Second, in this study, we measure firms' competitive strategy indirectly based on cluster analyses of R&D intensity, premium price capability, and asset efficiency. This might not represent the type of strategy that is actually pursued by the firm. Future research might obtain data regarding firms' pursued competitive strategy via surveys, and investigate whether the match between strategy goals and the types of executive compensation positively affects performance.

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Table 1

Sample Selection for Compustat S&P 500 Firms in Manufacturing Industry

Panel A: Selection of Firms:

1. Total number of firms listed in Compustat S&P 500	515
2. Less: Firms with incomplete data needed for cluster analyses	<u>271</u>
3. Total number of firms with complete data for cluster analyses	234
4. Initial numbers of firm-year observations	1,404
6. Less: Firm-years without complete data needed for data analyses	<u>655</u>
7. Total firm-year observations with complete data	<u>749</u>
	===

Panel B: Sampled firms according to their competitive strategy:

	<u>Number of firms</u>	<u>Firm-year observations</u>
Product differentiation strategy	84	258
Cost leadership strategy	<u>150</u>	<u>491</u>
Total	<u>234</u>	<u>749</u>
	===	===

Table 2
Business Strategy Derived From Hierarchical Cluster Analysis

Cluster Variables	Cluster 1 Product Differentiation ⁴	Cluster 2 Cost Leadership ⁴	Statistical Differences (t-test)
R&D Intensity ¹	0.0608	0.0260	8.159***
Asset Utilization Efficiency ²	0.6296	1.3068	-35.353***
Premium Price Capability ³	0.3952	0.3247	5.870***

¹ R&D intensity is measured by the ratio of research and development expenses to total sales revenues. A higher number indicates a more intensive investment in research and development. Since product differentiation firms are characterized by heavy investment in R&D, they are expected to score highly in this dimension.

² Asset Utilization Efficiency is measured by the ratio of total sales revenues to total assets. A higher number indicates a more efficient asset utilization. Since cost leadership firms try to offer the lowest prices, they are expected to score highly in this dimension.

³ Premium price capability is measured by the ratio of gross margin (total sales – cost of goods sold) to total sales revenue. A higher number indicates more premium prices charged to their customers. Since product differentiation firms are better able to charge premium prices to their customers, they are expected to score highly in this dimension.

⁴ The figures reported in these two columns are the actual means of the variables.

*** denotes a significance level of .001

Table 3
Descriptive Statistics^{a,b}

Variables	Mean	Standard Deviation	Minimum	Maximum	Units
PERFORM	40.44	225.54	-25.00	6,150.79	Percentage
PLC	75.68	12.65	29.16	97.86	Percentage
MSOWN	0.82	3.55	0.00	38.62	Percentage
PLTIP	8.88	11.71	0.00	72.42	Percentage
STRA	0.64	0.481	0.000	1.000	Code
STRA*PLC	48.66	36.84	0.00	66.96	-
STRA*MSOWN	0.49	1.90	0.00	21.36	-
STRA*:PLTIP	4.33	9.22	0.00	72.42	-
MTGS	7.44	2.63	2	30	Number of meetings
MTGSFEE	7.70	8.01	0.00	60.00	Thousands of US dollars
RETYRS	55.62	50.46	0	208	Number of years
SIZE	18,690.27	58,167.58	114.98	750,507.00	Millions of US dollars

^a The overall sample consists of 234 firms (84 PD firms and 150 CL firms) or 749 firm-year observations (258 firm-year observations for PD firms and 491 firm-year observations for CL firms).

^b PERFORM = one-year total return to shareholders, including the monthly reinvestment of dividends (ExecuComp data item TRS1YR). PLC = Performance-linked compensation of the firm determined by the percentage of incentive-based compensation to total compensation. MSOWN = Percentage of the firm's shares owned by the named executive officers. PLTIP = the amount paid out to the executive under the firm's long-term incentive plan as a percentage of total compensation. STRA = an indicator equal to 1 for cost leadership firms and 0 for product differentiation firms. STRA*PLC = the interaction between the STRA and PLC. STRA*MSOWN = the interaction between STRA and MSOWN. STRA*PLTIP = the interaction between STRA and PLTIP. MTGS = Number of board meetings held during the year. MTGSFEE = total amount paid for attending the board meetings. RETYRS = number of years of credited services the named executive officers have under the firm's pension plan. SIZE = firm size as measured by the logarithmic function of the firm's total assets.

Table 4

Pearson's (bottom) and Spearman's (top) correlations among variables (n=749) (p-values in parenthesis)^a

Variable	PERFORM	PLC	MSOWN	PLTIP	STRA	STRA* PLC	STRA* MSOWN	STRA* PLTIP	MTGS	MTGSFEE	RETYRS	SIZE
PERFORM	1	0.096 (0.006)	0.111 (0.002)	0.080 (0.023)	0.043 (0.158)	0.052 (0.092)	0.052 (0.154)	0.076 (0.014)	-0.129 (0.000)	-0.013 (0.712)	-0.035 (0.328)	-0.039 (0.269)
PLC	0.071 (0.044)	1	-0.273 (0.000)	0.252 (0.000)	0.120 (0.001)	0.294 (0.000)	-0.251 (0.000)	0.111 (0.002)	0.135 (0.000)	-0.145 (0.000)	0.106 (0.000)	0.555 (0.000)
MSOWN	0.103 (0.004)	-0.087 (0.016)	1	-0.273 (0.000)	0.031 (0.393)	0.013 (0.725)	0.836 (0.000)	0.034 (0.344)	-0.274 (0.000)	0.079 (0.029)	-0.139 (0.000)	-0.474 (0.000)
PLTIP	0.034 (0.335)	0.278 (0.000)	-0.033 (0.361)	1	0.115 (0.001)	0.134 (0.000)	-0.054 (0.138)	0.666 (0.000)	-0.081 (0.024)	-0.003 (0.927)	0.126 (0.000)	0.160 (0.000)
STRA	0.034 (0.335)	-0.147 (0.000)	-0.027 (0.451)	-0.009 (0.791)	1	0.944 (0.000)	0.229 (0.000)	0.657 (0.000)	-0.195 (0.000)	-0.066 (0.059)	0.205 (0.000)	0.060 (0.090)
STRA*PLC	-0.025 (0.474)	0.088 (0.012)	-0.038 (0.293)	0.067 (0.057)	0.960 (0.000)	1	0.167 (0.000)	0.669 (0.000)	-0.128 (0.000)	-0.095 (0.007)	0.092 (0.009)	-0.012 (0.734)
STRA*MSOWN	-0.006 (0.867)	-0.088 (0.015)	0.511 (0.000)	-0.062 (0.087)	0.187 (0.000)	0.158 (0.000)	1	0.167 (0.000)	-0.263 (0.000)	0.039 (0.277)	-0.080 (0.026)	-0.411 (0.000)
STRA*PLTIP	0.026 (0.402)	0.193 (0.000)	-0.048 (0.186)	0.724 (0.000)	0.478 (0.000)	0.479 (0.000)	0.012 (0.739)	1	-0.170 (0.000)	-0.041 (0.246)	0.092 (0.009)	-0.046 (0.196)
MTGS	-0.072 (0.042)	0.123 (0.000)	-0.049 (0.175)	-0.076 (0.032)	-0.180 (0.000)	-0.157 (0.000)	-0.112 (0.000)	-0.141 (0.000)	1	0.244 (0.000)	0.034 (0.337)	0.264 (0.000)
MTGSFEE	-0.005 (0.884)	-0.113 (0.000)	0.023 (0.517)	-0.008 (0.822)	-0.030 (0.398)	-0.066 (0.061)	-0.025 (0.486)	-0.049 (0.164)	0.244 (0.000)	1	-0.067 (0.056)	-0.138 (0.000)
RETYRS	-0.070 (0.047)	0.113 (0.001)	-0.104 (0.004)	0.126 (0.000)	0.011 (0.745)	0.046 (0.194)	-0.063 (0.079)	0.080 (0.023)	0.034 (0.337)	-0.067 (0.057)	1	0.390 (0.000)
SIZE	-0.085 (0.015)	0.523 (0.000)	-0.137 (0.000)	0.136 (0.000)	0.216 (0.000)	-0.163 (0.000)	-0.148 (0.000)	-0.006 (0.867)	0.247 (0.000)	-0.120 (0.001)	0.403 (0.000)	1

a PERFORM = one-year total return to shareholders, including the monthly reinvestment of dividends (ExecuComp data item TRS1YR). PLC = Performance-linked compensation of the firm determined by the percentage of incentive-based compensation to total compensation. MSOWN = Percentage of the firm's shares owned by the named executive officers. PLTIP = the amount paid out to the executive under the firm's long-term incentive plan as a percentage of total compensation. STRA = an indicator equal to 1 for cost leadership firms and 0 for product differentiation firms. STRA*PLC = the interaction between the STRA and PLC. STRA*MSOWN = the interaction between STRA and MSOWN. STRA*PLTIP = the interaction between STRA and PLTIP. MTGS = Number of board meetings held during the year. MTGSFEE = total amount paid for attending the board meetings. RETYRS = number of years of credited services the named executive officers have under the firm's pension plan. SIZE = firm size as measured by the logarithmic function of the firm's total assets.

Table 5
Regression Results of PERFORM on Business Strategy (STRA), Executive Compensation (PLC, MSOWN, LTIP), Interaction between Strategy and Executive Compensation (STRA*PLC, STRA*MSOWN, STRA*PLTIP), and Control Variables (MTGS, MTGSFEE, RETYRS, SIZE^a)

Variables (1)	Prediction (2)	Results					
		Regression 1 (with interaction terms)			Regression 2 (without interaction terms)		
		Coefficients (3)	z-values ^b (4)	VIF (5)	Coefficients (6)	t-values ^b (7)	VIF (8)
Intercept	?	-7.49	-0.23	-	31.11	1.41	-
PLC	+	1.71	3.79***	1.69	1.06	3.75***	1.62
MSOWN	+	3.04	3.31***	1.40	2.02	2.56***	1.02
PLTIP	+	-0.88	-2.14**	2.82	0.06	0.22	1.12
STRA	?	55.06	1.44	3.47	-12.02	-1.93*	1.13
STRA*PLC	-	-1.03	-2.03**	4.08	-	-	-
STRA*MSOWN	-	-3.28	-1.88**	1.43	-	-	-
STRA*PLTIP	-	1.49	2.90***	3.48	-	-	-
MTGS	+	-1.83	-1.47*	1.20	-1.72	-1.37*	1.19
MTGSFEE	+	0.08	0.22	1.11	0.12	0.33	1.11
RETYRS	+	0.01	0.20	1.28	-0.01	-0.04	1.28
SIZE	+	-8.10	-3.17***	2.39	-7.67	-3.01***	1.84
R ²			0.06			0.04	
Wald Chi ²			45.77***			31.82***	
Sample size			749			749	

^a PERFORM = one-year total return to shareholders, including the monthly reinvestment of dividends (ExecuComp data item TRS1YR). PLC = Performance-linked compensation of the firm determined by the percentage of incentive-based compensation to total compensation. MSOWN = Percentage of the firm's shares owned by the named executive officers. PLTIP = the amount paid out to the executive under the firm's long-term incentive plan as a percentage of total compensation. STRA = an indicator equal to 1 for cost leadership firms and 0 for product differentiation firms. STRA*PLC = the interaction between the STRA and PLC. STRA*MSOWN = the interaction between STRA and MSOWN. STRA*PLTIP = the interaction between STRA and PLTIP. MTGS = Number of board meetings held during the year. MTGSFEE = total amount paid for attending the board meetings. RETYRS = number of years of credited services the named executive officers have under the firm's pension plan. SIZE = firm size as measured by the total assets of the firm.

^b *, **, and *** denote the significant level of 0.10, 0.05 and 0.01 respectively based on two-tailed tests, except in cases of a directional prediction, in which a one-tailed test is used.

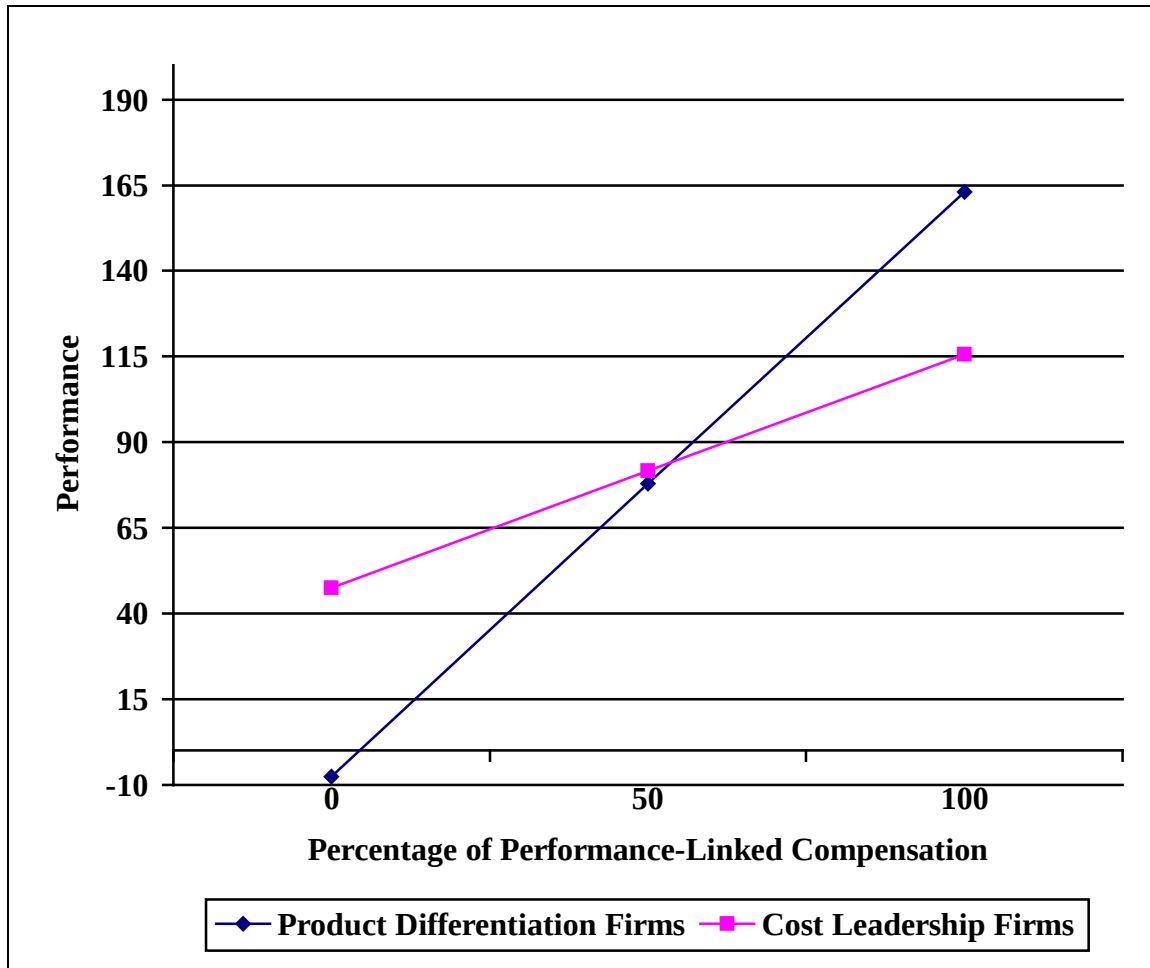
Table 6
Regression Results of PERFORM on Business Strategy (STRA) and Long-Term Incentive Plan (PLTIP)^a

Variables (1)	Prediction (2)	Coefficient (3)	z-values^b (4)
Panel A (Perform_{3years}):			
Intercept	?	2.78	1.09
PLTIP	+	0.45	3.09***
STRA	?	5.11	1.60
STRA*PLTIP	-	-0.18	-0.99
R ²			0.03
Wald Chi ²			17.31***
Sample size			791
Panel B (Perform_{5years}):			
Intercept	?	3.87	2.28**
PLTIP	+	0.22	2.65***
STRA	?	0.11	0.05
STRA*PLTIP	-	0.14	1.36
R ²			0.07
Wald Chi ²			39.68***
Sample size			759

^a PERFORM = three or five-year total return to shareholders, including the monthly reinvestment of dividends (ExecuComp data item TRS1YR). PLTIP = the amount paid out to the executive under the firm's long-term incentive plan as a percentage of total compensation. STRA = an indicator equal to 1 for cost leadership firms and 0 for product differentiation firms. STRA*PLTIP = the interaction between STRA and PLTIP.

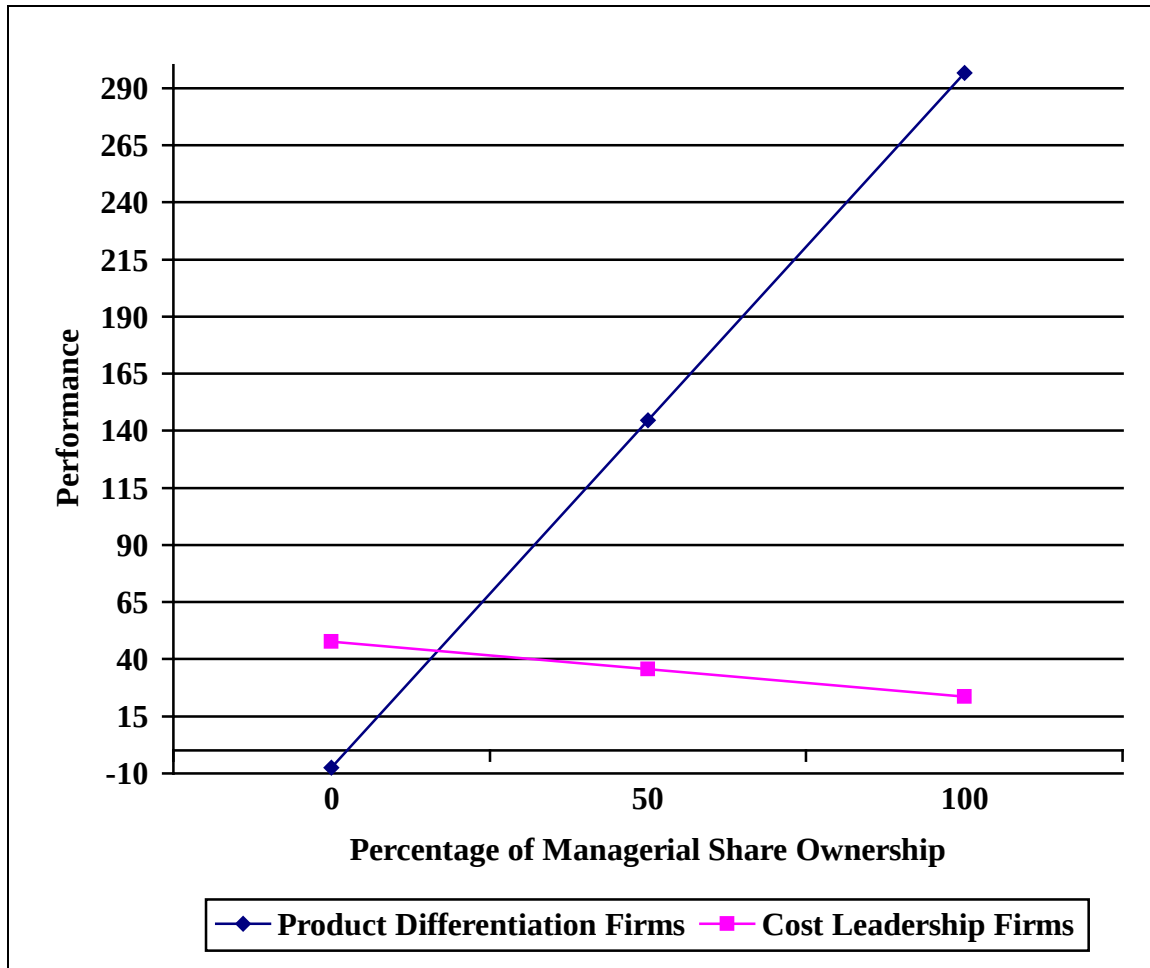
^b *, **, and *** denote the significant level of 0.10, 0.05 and 0.01 respectively based on two-tailed tests, except in cases of a directional prediction, in which a one-tailed test is used.

Figure 1
The Effects of Business Strategy on the Relationship between Performance-linked Compensation and Performance



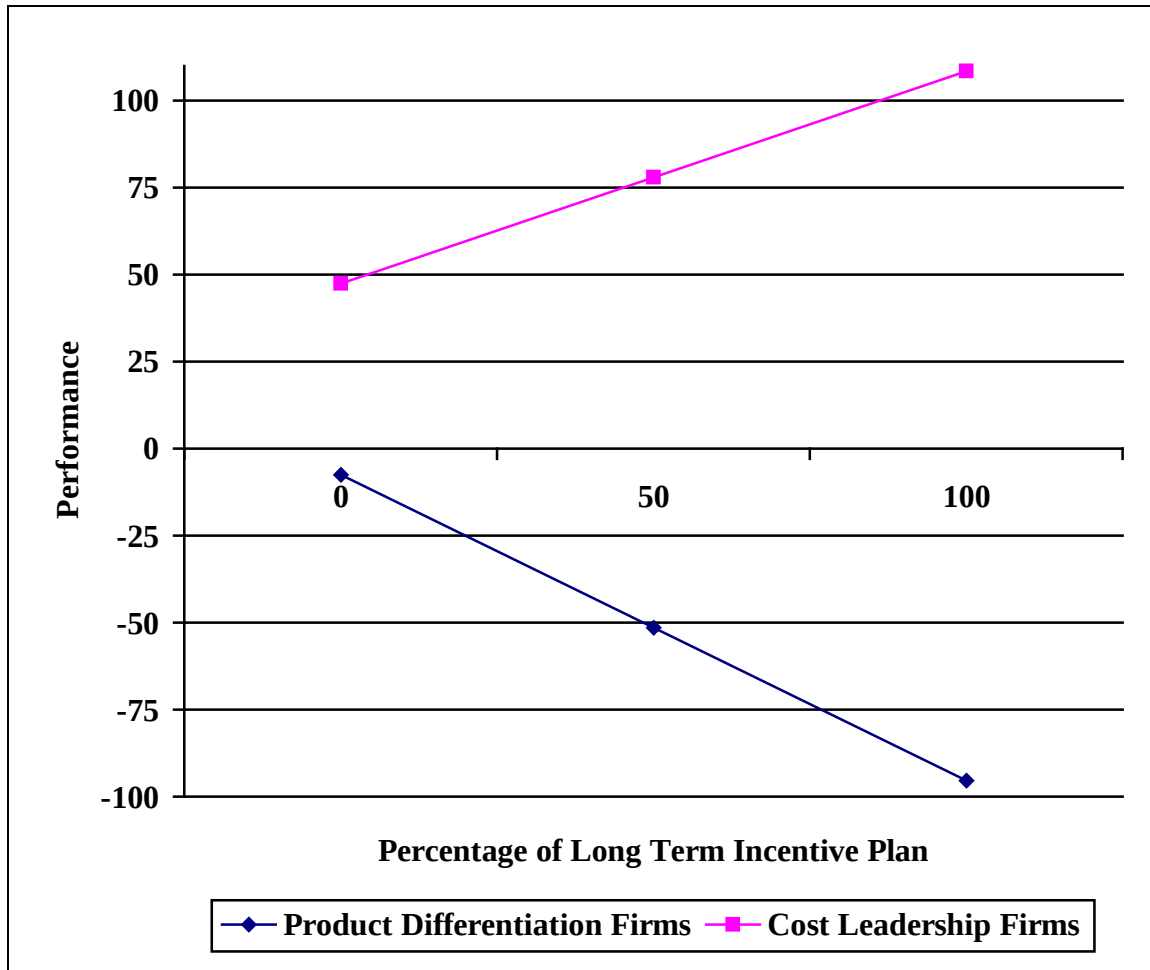
^b Conditioned on STRA being equal to 1 for cost leadership firms and 0 for product differentiation firms, the performance values for cost leadership firms are the sum of the estimated coefficients for the intercept, PLC, STRA, and STRA*PLC. The performance values for product differentiation firms are the sum of the estimated coefficients for the intercept and PLC.

Figure 2
The Effects of Business Strategy on the Relationship between Managerial Share Ownership and Performance



^b Conditioned on STRA being equal to 1 for cost leadership firms and 0 for product differentiation firms, the performance values for cost leadership firms are the sum of the estimated coefficients for the intercept, MSOWN, STRA, and STRA*MSOWN. The performance values for product differentiation firms are the sum of the estimated coefficients for the intercept and MSOWN.

Figure 3
The Effects of Business Strategy on the Relationship between Long-Term Incentive Plan and Performance



^b Conditioned on STRA being equal to 1 for cost leadership firms and 0 for product differentiation firms, the performance values for cost leadership firms are the sum of the estimated coefficients for the intercept, PLTIP, STRA, and STRA*PLTIP. The performance values for product differentiation firms are the sum of the estimated coefficients for the intercept and PLTIP.